

SOFTWARE ENGINEERING

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

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PROBLEM STATEMENT #63 | Sustainability & Green Tech

Community Solar Credit Allocation Manager

Board Key: CSCAM

1. Agile Backlog: Epics & User Stories (Deliverable 1)

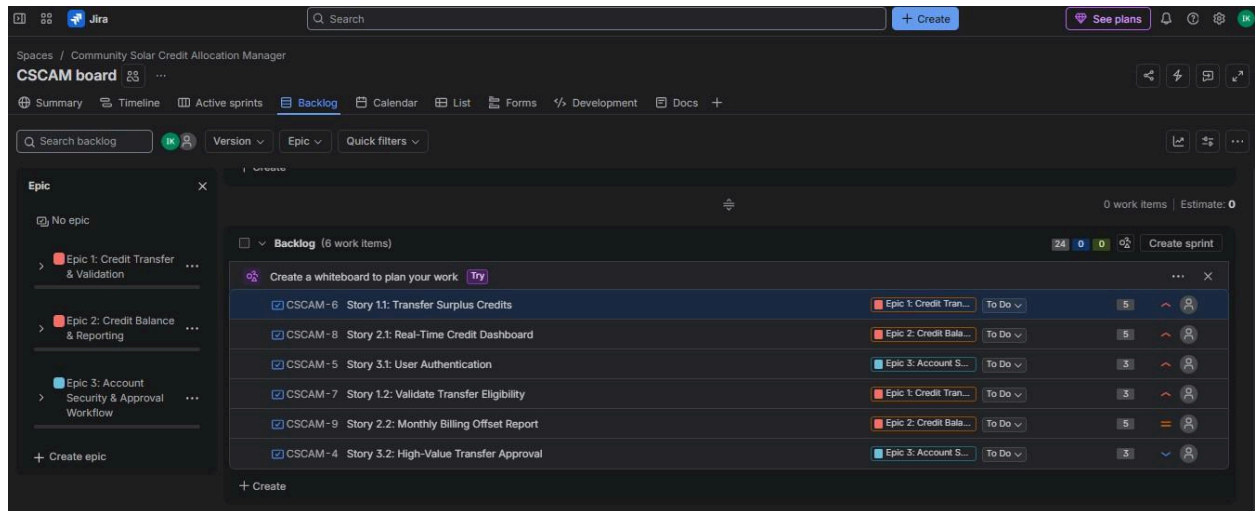
The functional requirements identified in Lab 1 (FR-001 through FR-005, plus the UC-02 "Transfer Surplus Credits" use-case flow) were grouped into three major functional themes (Epics) and broken down into six User Stories.

1.1 Epic Overview

Epic	Linked FRs	Description
Epic 1: Credit Transfer & Validation	FR-001	Enable prosumers to transfer surplus solar credits to neighbor accounts, with built-in validation of balance and recipient eligibility.
Epic 2: Credit Balance & Reporting	FR-002, FR-003	Give prosumers real-time visibility into their credit balance and generation history, and generate monthly billing offset reports for reconciliation.
Epic 3: Account Security & Approval Workflow	FR-004, FR-005	Secure portal access via authentication, and require Co-op Manager approval for any high-value (>500 kWh) credit transfers.

1.2 User Stories

ID	User Story	Epic	Priority
1.1	As a Prosumer Resident, I want to transfer my surplus solar credits to a neighbor's account, so that I can share my excess energy production and reduce their utility bill.	Epic 1	High
1.2	As a Prosumer Resident, I want the system to validate my transfer amount and recipient ID before confirming, so that I don't exceed my balance or send credits to an invalid account.	Epic 1	High
2.1	As a Prosumer Resident, I want to view my current solar credit balance and generation history on a dashboard, so that I can make informed decisions about transferring credits.	Epic 2	High
2.2	As a Co-op Manager, I want the system to generate a monthly report of utility billing offsets for each prosumer, so that billing reconciliation stays accurate and auditable.	Epic 2	Medium
3.1	As a Prosumer Resident or Co-op Manager, I want to securely log into the portal with my credentials, so that my energy and financial data stays protected.	Epic 3	High
3.2	As a Co-op Manager, I want to review and approve any credit transfer exceeding 500 kWh, so that I can prevent fraudulent or erroneous mass transfers.	Epic 3	Low

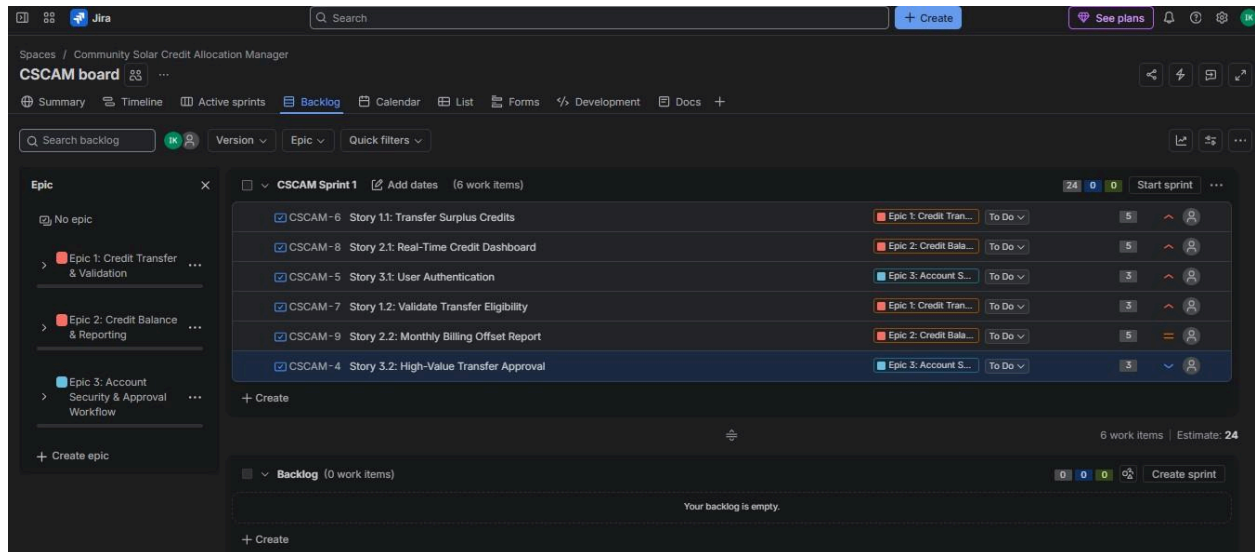


2. Story Point Assignments (Deliverable 2)

Story points were assigned using the Fibonacci scale (1, 2, 3, 5, 8...) based on estimated complexity, effort, and uncertainty, following a mental "planning poker" pass over each story.

ID	Story	Epic	Priority	Points
1.1	Transfer Surplus Credits	Credit Transfer & Validation	High	5
2.1	Real-Time Credit Dashboard	Credit Balance & Reporting	High	5
3.1	User Authentication	Account Security & Approval Workflow	High	3
1.2	Validate Transfer Eligibility	Credit Transfer & Validation	High	3
2.2	Monthly Billing Offset Report	Credit Balance & Reporting	Medium	5

3.2	High-Value Transfer Approval	Account Security & Approval Workflow	Low	3
	Total Sprint Estimate			24

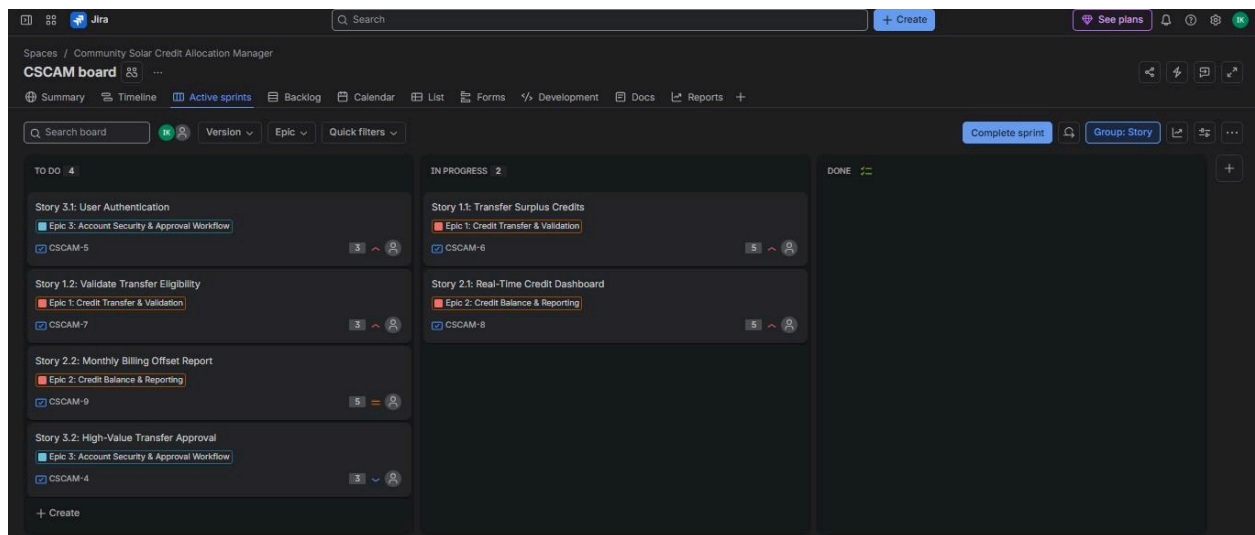
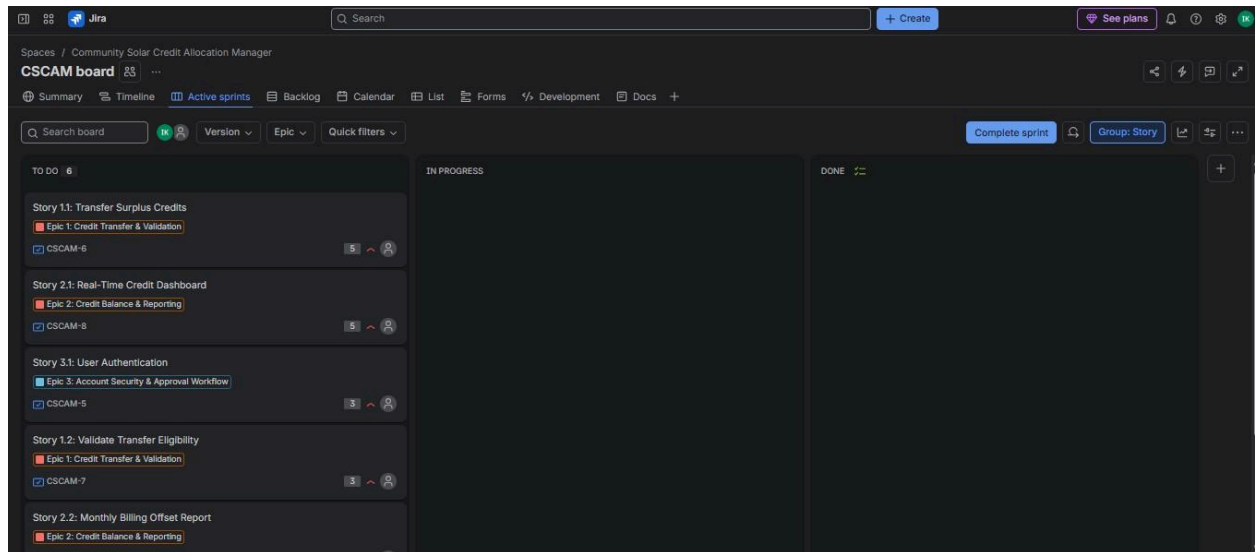


3. Sprint Simulation: Day-Wise Progress

Sprint 1 (1-week duration) was started with all 6 stories, ranked by priority. Rather than moving every story to Done in a single sitting, progress was simulated across four days to reflect a realistic sprint cadence, deliberately staggering completions so the Burndown Chart shows a genuine step-wise decline rather than a single flat drop.

Day 1 - Sprint Start

Sprint 1 was started with all 6 stories (Estimate: 24 points) sitting in the To Do column. The two highest-priority, highest-effort stories - Story 1.1 (Transfer Surplus Credits) and Story 2.1 (Real-Time Credit Dashboard), both High priority / 5 points - were moved into In Progress to kick off work on the core transfer and dashboard features first.

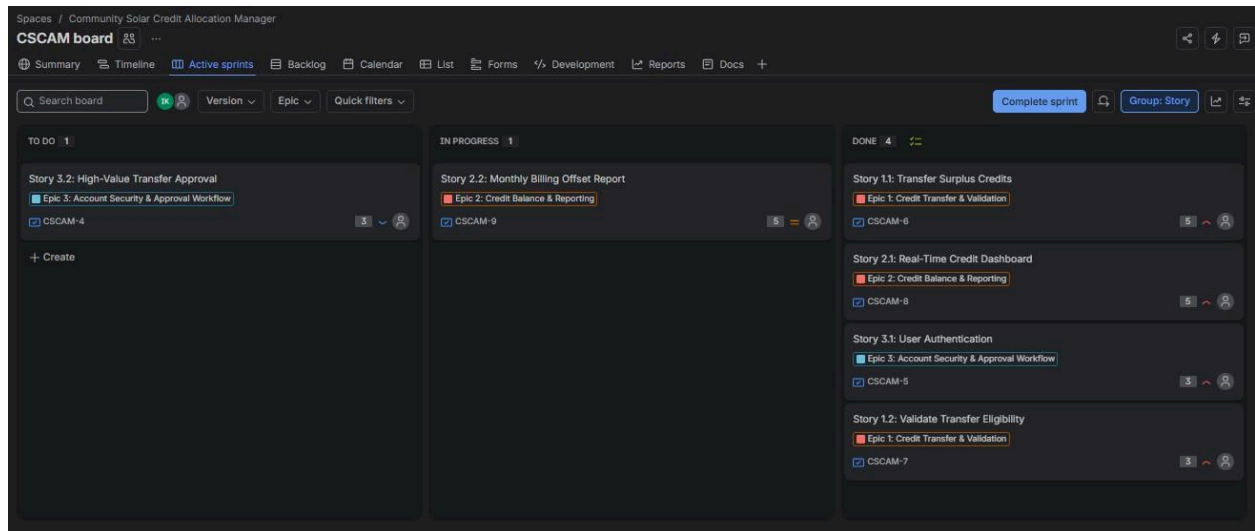


Day 2 - First Completions

Story 1.1 and Story 2.1 (10 points combined) were marked Done, and the next two High-priority stories , Story 3.1 (User Authentication) and Story 1.2 (Validate Transfer Eligibility), 3 points each - were moved into In Progress. This produced the first visible step down in the Burndown Chart, from 24 remaining points to approximately 14.

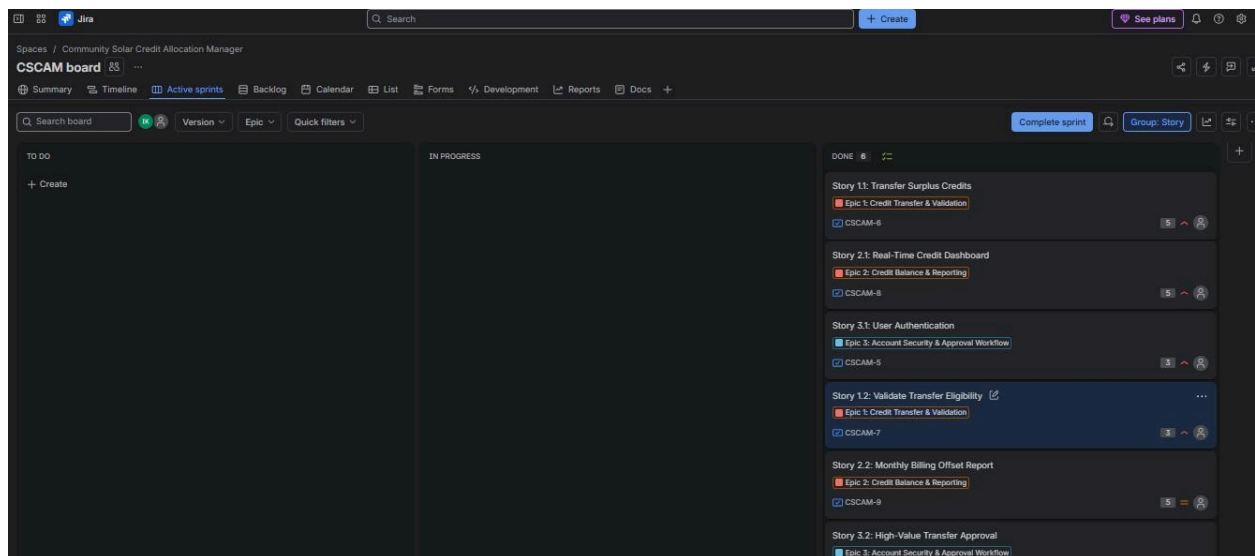
Day 3 - Mid-Sprint Progress

Story 3.1 and Story 1.2 (6 points combined) were completed, bringing 4 of 6 stories to Done. The remaining Medium-priority story, Story 2.2 (Monthly Billing Offset Report, 5 points), was moved into In Progress, leaving only Story 3.2 (High-Value Transfer Approval, Low priority, 3 points) in To Do. Remaining work dropped to 8 points on the Burndown Chart at this stage.



Day 4 - Sprint Completion

Story 2.2 was completed, followed by Story 3.2, bringing all 6 stories to Done and remaining work down to 0 points. With every story in the sprint completed, the sprint was ready to be closed via "Complete sprint."

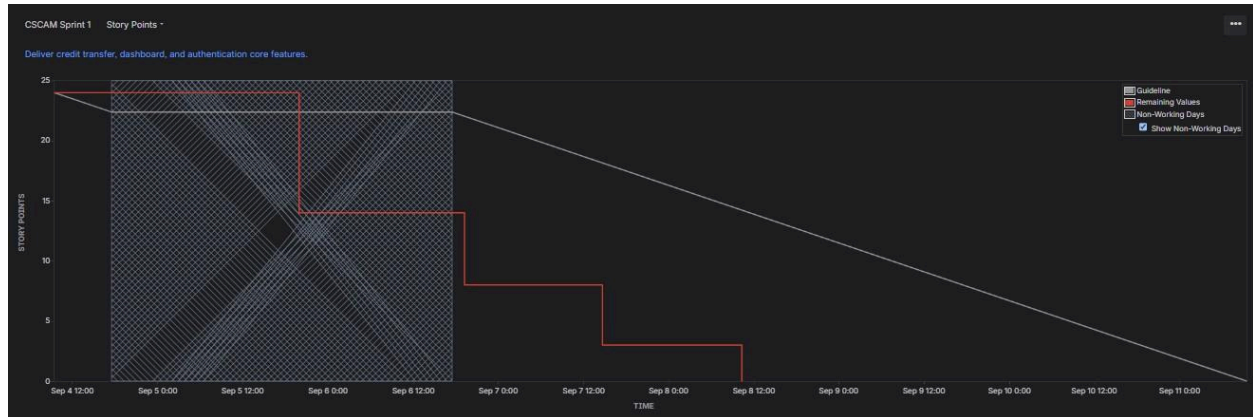


4. Active Sprint Board (Deliverable 3)

The Active Sprint view below shows the board at Day 4, fully progressed — all 6 stories in the Done column, with the Kanban-style To Do → In Progress → Done flow reflected across the full sprint duration.

5. Burndown Chart (Deliverable 4)

The Burndown Chart for CSCAM Sprint 1 tracks Story Points remaining against the ideal linear Guideline over the sprint duration (Sep 4 – Sep 11).



Reading the chart:

- The **grey Guideline** shows the ideal linear pace assuming steady daily progress across the full week.
- The **red Remaining Values line** shows actual progress: it stays flat at 24 through Day 1, then drops in four distinct steps to ~14 (Day 2, after Story 1.1 + 2.1), to ~8 (Day 3, after Story 3.1 + 1.2), and finally to 0 (Day 4, after Story 2.2 + 3.2).
- Because all work was completed by Sep 8 - well before the Sep 11 sprint end date - the red line finishes **below and ahead of** the grey Guideline, indicating the team closed the sprint early relative to the planned pace.

Reflection Questions

1. Did your estimations reflect the actual effort?

Yes. The Fibonacci-based estimates (5 points for the two most complex stories - Transfer Surplus Credits and Real-Time Credit Dashboard, since both involve backend balance logic and real-time data ingestion down to 3 points for simpler stories like Authentication and Validate Transfer Eligibility) held up well during the simulated sprint. No story required re-estimation mid-sprint, and the relative sizing ($5 > 3$) matched the relative complexity of implementing live smart-meter data feeds and dual-account credit transfers versus more contained features like login validation.

2. Was your backlog well-prioritized?

Yes. All four High-priority stories (1.1, 1.2, 2.1, 3.1) map directly to core FR-001, FR-002, and FR-005 - the requirements marked High priority in the original Lab 1 requirements table - and were correctly ranked above the Medium-priority reporting story (2.2, FR-003) and the Low-priority approval workflow (3.2, FR-004). This meant the highest-value, highest-risk functionality (credit transfer and authentication) was tackled first, which is consistent with good Agile prioritization practice of front-loading the riskiest or most valuable work.

3. How did your simulated sprint align with your plan?

The sprint aligned closely with the plan, with all 6 stories completed within 4 days of a 7-day sprint window rather than requiring the full week. Progress was staged deliberately - 2 stories per day into Done rather than completing everything at once, which meant the sprint didn't experience any unplanned blockers or the need to descope work. The one deviation from a purely linear pace was that Day 1 involved no completions (only starting 2 stories), which is realistic since the highest-effort stories (5 points each) naturally take longer to move from In Progress to Done than smaller ones.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart shows the team finishing all 24 story points a few days ahead of the 1-week sprint boundary, with the red Remaining Values line dropping below the grey Guideline from Day 2 onward.

Per the burndown chart's own interpretation guidance, consistently finishing early can be a sign the team under-committed during sprint planning, meaning a future sprint of similar duration could likely absorb additional stories (ex: pulling in 1–2 more backlog items) without risking an incomplete sprint. The step-wise shape of the line (rather than one flat diagonal) also reflects that work was completed in batches tied to related functionality (transfer stories together, then auth stories, then reporting/approval), rather than a perfectly even daily cadence which is normal for small, tightly-scoped backlogs like this one.